

3-Mark Questions & Answers -- DBMS Module 4: Relational Database Design

Q1. Compare 3NF and BCNF with a suitable example.

Property	3NF	BCNF
FD Condition $X \rightarrow Y$	X is Super Key OR Y is Prime Attri	X MUST be a Super Key (No exceptions).
Overlapping Keys	Allows overlapping candidate key depen	Eliminates overlapping key redundancy.
Dependency Preservation	Always Guaranteed.	Not always guaranteed.

* **Example:** ADVISOR(Student_ID, Subject, Advisor_Name) where Advisor_Name $\rightarrow$ Subject. It is in 3NF (Subject is prime) but NOT in BCNF (Advisor_Name is not a superkey).

Q2. Explain the algorithm for attribute closure X^+ .

Input: Set of attributes X, Set of FDs F
Output: X^+ (Attribute Closure)

- Set $X^+ = X$
- Repeat until X^+ does not change:
For each FD $(Y \rightarrow Z)$ in F:
If Y is a subset of X^+ :
 $X^+ = X^+ \cup Z$
- Return X^+

Q3. Differentiate between functional dependency and multivalued dependency.

Feature	Functional Dependency (FD)	Multivalued Dependency (MVD)
Symbol	$X \rightarrow Y$	$X \twoheadrightarrow Y$
Mapping	Single value: X determines a unique si	Multiple values: X independent multide
Normal Form Target	2NF, 3NF, BCNF	4NF

Q4. What are NoSQL data models List its 4 categories.

NoSQL databases are non-relational, schema-less, horizontally scalable systems built for big data.

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Q5. Explain dependency preservation in database decomposition.

A decomposition of R into sub-relations $R_1, R_2, \dots, R_n$ is dependency preserving if the closure of the union of all functional dependencies enforced locally on each sub-relation equals the closure of the original set of functional dependencies:

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It ensures database integrity constraints can be enforced without expensive JOIN operations across tables.